Yiwen Lu
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Research Interests

My research centers on **self-improving and trustworthy LLM agents**. I study how agents turn knowledge, experience, and feedback into reliable actions, with particular interest in **opioid misuse prevention and scam detection**.

- **Agentic Post-Training and Decision-Making:** Training agents to act on expert feedback and verifiable outcomes in public-health decision support.
- **Experience Memory and Continual Adaptation:** Learning from past interactions to personalize opioid-prevention support and adapt to evolving scam tactics.
- **Trustworthy Long-Horizon Agents:** Failure-aware reasoning, robust tool use, and human oversight in multi-step anti-scam and public-health workflows.

Education

University of Notre Dame, Notre Dame, IN — Aug. 2025 – Present
Ph.D. student in Electrical Engineering; qualifier passed Aug. 2026; M.S. expected Dec. 2026.

Beijing Jiaotong University, Beijing, China — Sept. 2021 – Jul. 2025
B.S. in Computer Science.

Publications and Manuscripts

1. **SkillRC: Is It the Skill or the Stack? Auditing Experience Memory in Agent Harnesses**
   Yiwen Lu. *Manuscript, 2026* PDF Code
2. **Answer Supervision Teaches a Search Agent to Resist Wrong Instructions, Far More Than Wrong Claims**
   Yiwen Lu. *Manuscript, 2026* PDF Code
3. **Know2Act: When Does an Agent's Written Prediction Control Its Next Action?**
   Yiwen Lu. *Manuscript, 2026* PDF Code
4. **Ultra-Wideband Indoor Localization and Bistatic Deep Learning-Based Mapping for ISAC**
   Yiwen Lu, Mikhail Saleev, Toros Arikan *Under review at IEEE ICC 2027, SAC – Integrated Sensing and Communications, 2026*
5. **Deep Learning Estimation of the Number and Locations of Reflective Surfaces for Underwater ISAC**
   Yiwen Lu, Mikhail Saleev, Yitong Luo, Allison Sproul, Rene Alzina, Toros Arikan *Under review at IEEE ICASSP 2027, 2026* PDF
6. **Mobility-LLM: Learning Visiting Intentions and Travel Preferences from Human Mobility Data with Large Language Models** PDF
   Letian Gong, Yan Lin, Xinyue Zhang, **Yiwen Lu**, Xuedi Han, Yichen Liu, Shengnan Guo, Youfang Lin, Huaiyu Wan ***NeurIPS 2024***

Research Experience

University of Notre Dame — Independent Researcher — 2026 – Present
Reliability and Adaptation in LLM Agent Systems

- Inspired by discussions with Notre Dame peers, independently examined how learned predictions, experience memory, and training feedback shape decisions in tool-using LLM agents.
- **SkillRC:** Built a matched-control evaluation of experience memory across six ALFWorld readers and WebShop; found no measurable ALFWorld reading cost, a 3.5-point gain for experience-derived versus equal-length generic rules, and negative content value on WebShop.
- **Answer Supervision Teaches a Search Agent:** Evaluated GRPO-trained agents under retrieved injections using a hash-committed sealed-test protocol; answer supervision reduced adoption of wrong instructions from 68.3% to 33.5%, compared with 20.7% to 17.7% for equivalent factual claims.
- **Know2Act:** Built a counterfactual ALFWorld study of knowledge-to-action causality; replacing written predictions redirected 80% of first actions after decision-consistent fine-tuning, while deleting predictions had little effect and 25% contradictory supervision removed this control.

University of Notre Dame — Graduate Research Assistant — Aug. 2025 – Present
Deep Learning for Spatial Sensing and Environmental Mapping

- **Indoor Localization and Bistatic Mapping:** Investigated ultra-wideband localization and deep-learning-based bistatic environmental mapping for integrated sensing and communications (ISAC); first-author manuscript under review at IEEE ICC 2027.
- **Underwater Scene Reconstruction:** Developed a multi-stage deep-learning and signal-processing pipeline for estimating reflective boundaries from noisy acoustic measurements, incorporating failure diagnosis and correction; first-author manuscript under review at IEEE ICASSP 2027.

Beijing Jiaotong University — Research Assistant — Apr. 2023 – Jun. 2025
LLM Adaptation for Spatiotemporal Intelligence | Advisors: Huaiyu Wan and Shengnan Guo

- Co-authored **Mobility-LLM (NeurIPS 2024)**, adapting pretrained LLMs to structured human mobility data for prediction tasks over heterogeneous spatiotemporal sequences.
- Implemented the **Visiting Intention Memory Network (VIMN)** and **Human Travel Preference Prompts (HTPP)**; evaluated the framework across four datasets and three downstream tasks.

Technical Skills

Programming and frameworks: Python, PyTorch, vLLM, Hugging Face Transformers, Search-R1; C/C++, Java, SQL, Bash.
Research methods: LLM post-training (SFT, GRPO), agent harnesses, experience memory, retrieval/tool-use evaluation, pre-registration, paired bootstrap, failure analysis, spatiotemporal and sensor-data modeling.